Anomaly Detection for Fraud Prevention

Abstract

This project addresses the critical challenge of fraud prevention through anomaly detection, with a special focus on the financial and insurance industries where identifying fraudulent activities is paramount. Traditional supervised learning methods often fall short due to the rarity of fraud cases—sometimes as few as one in a thousand or million transactions—leading to imbalanced datasets and an unacceptably high rate of false positives. To tackle this, we propose a practical approach using unsupervised clustering algorithms to establish a baseline of normal transaction behavior and flag deviations as potential fraud. Our research will explore cutting-edge unsupervised techniques, analyze transaction features to distinguish normal from abnormal patterns, and develop a clustering-based anomaly detection system. The solution will be rigorously tested using widely recognized datasets, such as the Kaggle credit card fraud dataset, and evaluated against industry standards to ensure reliability and effectiveness.

Implementation Plan

- **Data Preparation**: We will gather and carefully prepare datasets, including the banksim data and other fraud detection dataset, to extract meaningful transaction features that support our analysis.
- **Clustering Model Development**: Our team will apply unsupervised clustering algorithms to create a robust model that accurately reflects typical transaction patterns.
- **Anomaly Detection**: We will design a user-friendly mechanism to identify outliers as potential fraud cases, leveraging the insights from our clustering model.
- **Evaluation**: The performance of our solution will be thoroughly assessed using established metrics, with a comparative analysis against leading fraud detection systems to highlight its strengths.
- **Documentation**: We will compile a comprehensive report of our findings, offering actionable insights to support the practical application of this system in fraud prevention efforts.